Compare all Ansible, Puppet, and Chef parameters.

Ansible, Puppet, and Chef are all popular configuration management tools, each with its own set of features and parameters. Here

No.	Ansible	Puppet	Chef
	ansible • **Purpose**: Run ad-hoc commands on remote hosts. • **Example**: ansible all -m ping • **Parameters**: -i: Inventory file (ansible -i inventory.ini all -m ping) -u: Remote user (ansible -u ubuntu all -m ping) -k: Prompt for SSH password	**puppet apply** • **Purpose**: Apply a Puppet manifest on the local system. • **Example**: puppet apply site.pp • **Parameters**: --modulepath: Specify module path (puppet apply --modulepath=modules site.pp) --noop: Run in no-operation mode (puppet apply --noop site.pp) --debug: Enable debug logging (puppet apply --debug site.pp)	**chef-client** • **Purpose**: Run the Chef client to apply configurations from a Chef server. • **Example**: chef-client • **Parameters**: -c: Specify the configuration file (chef-client -c /path/to/chef-client.rb) -j: Specify the JSON file for node attributes (chef-client -j node.json) -E: Specify the environment (chef-client -E production)
	ansible-playbook • **Purpose**: Run playbooks to configure or manage systems. • **Example**: ansible-playbook site.yml • **Parameters**: -i: Inventory file (ansible-playbook -i inventory.ini site.yml) -u: Remote user (ansible-playbook -u ubuntu	**puppet agent** • **Purpose**: Run the Puppet agent to apply configurations from a Puppet master. • **Example**: puppet agent -t • **Parameters**: -t: Run in test mode (puppet agent -t) --certname: Specify the certificate name (puppet agent --certname mycertname -t)	**knife** • **Purpose**: A command-line tool for interacting with the Chef server. • **Example**: knife cookbook upload my_cookbook • **Parameters**: cookbook upload: Upload a cookbook (knife cookbook upload <cookbook_name>) node list: List nodes (knife node list) role list: List roles (knife role list) data bag list: List data bags (knife data bag list)

	site.yml) -k: Prompt for SSH password --tags: Run specific tasks (ansible-playbook site.yml --tags "install") --extra-vars: Pass extra variables (ansible-playbook site.yml --extra-vars "version=2.0")	--onetime: Run only once (puppet agent --onetime)	
	ansible-galaxy • **Purpose**: Manage roles from the Ansible Galaxy repository. • **Example**: ansible-galaxy install geerlingguy.apache • **Parameters**: install: Install roles (ansible-galaxy install <role_name>) remove: Remove roles (ansible-galaxy remove <role_name>) list: List installed roles (ansible-galaxy list)	**puppet cert** • **Purpose**: Manage certificates for secure communication. • **Example**: puppet cert list • **Parameters**: list: List certificates (puppet cert list) sign: Sign a certificate request (puppet cert sign <certname>) revoke: Revoke a certificate (puppet cert revoke <certname>)	**chef-solo** • **Purpose**: Run Chef configurations locally without a Chef server. • **Example**: chef-solo -c solo.rb -j node.json • **Parameters**: -c: Specify the configuration file (chef-solo -c solo.rb) -j: Specify the JSON file for node attributes (chef-solo -j node.json) -o: Specify a run list (chef-solo -o "recipe[my_cookbook::default]")

	ansible-vault • **Purpose**: Manage encrypted files. • **Example**: ansible-vault encrypt secrets.yml • **Parameters**: encrypt: Encrypt files (ansible-vault encrypt secrets.yml) decrypt: Decrypt files (ansible-vault decrypt secrets.yml) edit: Edit encrypted files (ansible-vault edit secrets.yml)	**puppet module** • **Purpose**: Manage Puppet modules. • **Example**: puppet module install puppetlabs-ntp • **Parameters**: install: Install a module (puppet module install <module_name>) remove: Remove a module (puppet module uninstall <module_name>) list: List installed modules (puppet module list)	**chef-shell** • **Purpose**: Open an interactive Ruby shell with Chef-specific extensions. • **Example**: chef-shell • **Parameters**: --no-plugins: Disable plugins (chef-shell --no-plugins) --irb: Start an IRB session (chef-shell --irb)

Summary

- **Ansible**: Uses simple commands for ad-hoc tasks, running playbooks, managing roles, and encrypted files. It is agentless and operates over SSH.
- **Puppet**: Uses commands to apply manifests, manage agents, handle certificates, and manage modules. It operates using a master-agent model.
- **Chef**: Uses commands to run client configurations, manage cookbooks, nodes, roles, and data bags. It can run in both client-server and standalone modes.

Each tool has a unique approach to configuration management, and the choice depends on your specific needs and preferences.